Cardiovascular System (CARD) · NPLEX-P1-CARD-15

CV-15 Heart and Lung Interactions

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-15-01-C01				
NPLEX-P1-CARD-15-01-C02				
NPLEX-P1-CARD-15-01-C03				
NPLEX-P1-CARD-15-01-C04				
NPLEX-P1-CARD-15-01-C05				
NPLEX-P1-CARD-15-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C01-BD1

Build the two circuits side by side

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C01 Activity: NPLEX-P1-CARD-15-01-C01-BD1 Date: _____

Without looking at your notes, draw the pulmonary and systemic circuits in series and write on the drawing the resting mean arterial pressure, the receiving atrial pressure, the capillary hydrostatic pressure, and the resistance in Wood units for each circuit. Beside the drawing write the flow each ventricle carries at rest, and show from pressure and flow how the resistance figures follow. Name the main site of resistance in each circuit and write two sentences on the wall thickness and smooth muscle differences that explain it. Then write the volume of blood the pulmonary circuit holds, what supplies the airways themselves and at what pressure and what share of cardiac output, and the two consequences of running the lung at low pressure. Finish with the mean pulmonary artery pressure that defines pulmonary hypertension.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C01-BD2

Work the pulmonary resistance numbers through exercise and disease

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C01 Activity: NPLEX-P1-CARD-15-01-C01-BD2 Date: _____

A person goes from rest, with a cardiac output of 5 L/min, a mean pulmonary artery pressure of 15 mmHg, and a left atrial pressure of 5 mmHg, to heavy exercise, with a cardiac output of 15 L/min, a mean pulmonary artery pressure of 22 mmHg, and a left atrial pressure of 10 mmHg. Calculate pulmonary vascular resistance at rest and in exercise, show your working, and write what the mean pulmonary artery pressure would have been in exercise had resistance not changed. Name the two passive mechanisms that lowered it, say which one dominates at low pressure and which at higher pressure, and explain why a small change in vessel radius matters so much. Write what happens to capillary transit time and why gas exchange still completes. Then write what pulmonary vascular resistance does at high lung volume, at low lung volume, and near functional residual capacity, naming the class of vessel responsible in each case. Finish with two patients, one after removal of a healthy lung and one with half the vascular bed lost to emphysema and old emboli: write whose resting pulmonary artery pressure rises, whose rises steeply on exercise, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C02-BD1

Draw hypoxic pulmonary vasoconstriction from sensor to constriction

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C02 Activity: NPLEX-P1-CARD-15-01-C02-BD1 Date: _____

Without your notes, draw a small pulmonary artery supplying a poorly ventilated alveolus and write out the chain of events from low alveolar oxygen to contraction of that vessel, one step per line. Name the channel family that is inhibited first, what that does to the membrane potential, which calcium channel opens next, and the two routes by which calcium reaches the contractile proteins. Add the enzyme that raises calcium sensitivity and name the two endothelial mediators that shift in this setting and in which direction each moves. Write which gas tension is the stimulus, the value below which the response becomes strong, and the evidence from an isolated denervated lung that the response is local. Then write the purpose this response serves and what the leading proposal is for the upstream oxygen sensor.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C02-BD2

Regional against global hypoxia, and what chronic hypoxia leaves behind

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C02 Activity: NPLEX-P1-CARD-15-01-C02-BD2 Date: _____

Three people are hypoxic: one with right lower lobe pneumonia, one who has climbed rapidly to 4,000 m, and one with twenty years of COPD and chronic hypoventilation. For each, write whether the hypoxia is regional or global, what hypoxic pulmonary vasoconstriction achieves, what happens to mean pulmonary artery pressure, and what happens to arterial oxygen. Explain why the same response helps in one case and harms in another. Write what hypercapnia and acidosis do to the constriction. For the climber, explain the accepted mechanism of high-altitude pulmonary edema from uneven constriction. For the patient with COPD, write the transcription factors stabilised by sustained hypoxia, the three structural changes they drive in the vessel wall, the hormone they raise and what the resulting blood change does to resistance, and which part of the pulmonary hypertension no longer reverses with oxygen. Finish with the fetus: write what keeps pulmonary vascular resistance high before birth, where right ventricular output goes instead, the three changes at birth that drop resistance, and the name for failure of that drop.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-15-01-C02-RT01

Rebuild the hypoxic pulmonary vasoconstriction sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C02 Activity: NPLEX-P1-CARD-15-01-C02-RT01 Date: _____

Rebuild the six-step sequence that runs from a fall in regional alveolar oxygen to blood being diverted toward ventilated lung, one step per line and in order. Beside the first step write the alveolar oxygen tension at which the response becomes strong. Beside the channel step write which channels are involved and what holds the resting potential negative when they are open. Beside the calcium step write the two sources of calcium and the enzyme that raises calcium sensitivity. Underneath, draw the same vessel in a systemic tissue bed made hypoxic and write beside it the channel that opens there, the direction the membrane potential moves, and the resulting change in vessel diameter.

Rebuild from memory: Hypoxic pulmonary vasoconstriction

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-15-01-C02-CP01

Pulmonary against systemic arterioles

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Pulmonary arterioles	Systemic arterioles
Response to local hypoxia		
Key ion channel event and the direction the membrane potential moves		
Other mediators that contribute to the response		
What the response accomplishes for the body		
Response to acidosis and a high carbon dioxide tension		
Result when hypoxia involves the whole bed at once		
Amount of smooth muscle in the wall and how much of the circuit's resistance sits there		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C03-BD1

One breath, four chambers, and the second heart sound

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C03 Activity: NPLEX-P1-CARD-15-01-C03-BD1 Date: _____

Write the definition of transmural pressure and why a change in pleural pressure changes the filling of every cardiac chamber at once. Give pleural pressure at functional residual capacity and at the end of a quiet inspiration. Write the two pressures whose difference drives venous return and the normal value of the upstream one. Then take one spontaneous inspiration and write in order what happens to right atrial pressure, abdominal pressure, venous return, and right ventricular stroke volume. List the three reasons left ventricular stroke volume falls slightly in the same breath, and write when the extra right ventricular output reaches the left heart. Draw one normal second heart sound in inspiration and one in expiration, label A2 and P2, and write beside the inspiratory pair the two reasons P2 is delayed and the reason A2 is slightly early. Give the normal limit for the inspiratory fall in systolic pressure and where on the chest the split is best heard.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C03-BD2

Exaggerated and reversed breathing effects

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C03 Activity: NPLEX-P1-CARD-15-01-C03-BD2 Date: _____

Four patients are examined. The first has cardiac tamponade, the second a severe asthma attack, the third constrictive pericarditis, and the fourth is on a ventilator with a positive end-expiratory pressure of 15 cmH2O after major blood loss. For the first two, write the mechanism by which the inspiratory fall in systolic pressure exceeds 10 mmHg, naming for each whether the pericardium or the pleural pressure swing is doing the work. For the third, write why this finding is less common and what happens to the jugular venous pressure on inspiration instead. For the fourth, write what happens to right atrial pressure, to the gradient for venous return, to cardiac output, and to left ventricular afterload, and explain why the fall in output is larger because of the blood loss. Then write what happens in the same ventilated patient if the left ventricle is instead failing with high filling pressures, and why stroke volume may hold up. Finish by writing which phase of a Valsalva manoeuvre reproduces this pattern and what the heart rate does and through which reflex. Give the four S2 splitting patterns with the timing fault behind each and one example condition for each, including why the split in an atrial septal defect does not move with breathing.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-15-01-C03-RT01

Rebuild the inspiration sequence through to S2

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C03 Activity: NPLEX-P1-CARD-15-01-C03-RT01 Date: _____

Rebuild the six-step sequence that runs from a fall in pleural pressure to a split second heart sound and a small dip in arterial pressure, one step per line and in order. Beside the venous return step write the two forces that push blood toward the thorax. Beside the step where left stroke volume falls, write the three reasons it falls. Beside the final step write the normal limit for the fall in systolic pressure. Then write the same six steps as they run during expiration, showing that each one reverses, and mark on which phase of the breath the extra right ventricular output finally reaches the aorta.

Rebuild from memory: Inspiration, ventricular output, and S2

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C04-BD1

Starling forces at the pulmonary capillary and the safety margin

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C04 Activity: NPLEX-P1-CARD-15-01-C04-BD1 Date: _____

Draw a pulmonary capillary and its alveolus and mark every Starling force with its value and its direction: capillary hydrostatic pressure, interstitial hydrostatic pressure, interstitial oncotic pressure, and plasma oncotic pressure. Total the outward forces and the inward force, write the net filtration pressure, and write what carries that filtrate away. Then write the three things that hold off edema as capillary pressure climbs, the capillary pressure at which alveolar flooding begins, and the size of the safety margin in mmHg. Write in order what fills first and what floods second. Finish with three modifiers: what a low plasma albumin does to the margin, what years of raised pressure do to it and by how much the lymphatics can enlarge, and how permeability injury produces edema at a normal capillary pressure and how the fluid itself differs.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C04-BD2

Which failing chamber, and which side of the circuit caused it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C04 Activity: NPLEX-P1-CARD-15-01-C04-BD2 Date: _____

Two patients have raised pulmonary artery pressure and a failing right ventricle. One has twenty years of COPD and the other has long-standing mitral stenosis. For each, write the pulmonary hypertension group, the mechanism raising pulmonary artery pressure, whether the term cor pulmonale applies, and your reason. Write the five groups with their causes and say which ones load the right ventricle from the lung side. Then for the patient with COPD write how the right ventricle first adapts to slowly rising afterload and why a thicker wall helps, what happens to the tricuspid valve as it later dilates, the three systemic signs of the raised venous pressure, the two findings on examination that point to pulmonary hypertension, and the three ECG changes. For the patient with mitral stenosis write which measured pressure rises and its normal range, what shortens diastole and makes her worse, why the measured wedge pressure sits above left ventricular diastolic pressure here, the receptor whose stimulation causes her dyspnoea, why lying flat is worse, and the two chest film findings. Finish with a third patient who has a massive pulmonary embolus: write the two mechanisms obstructing the bed, the highest mean pressure a previously normal right ventricle can generate, and the share of the circulation whose obstruction brings acute failure or sudden death.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-15-01-C04-PR01

Predict the course of a massive pulmonary embolus

A large clot lodges in both main pulmonary arteries of a patient whose right ventricle was previously normal. Predict each item below, write the direction of the change, and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to right ventricular afterload, and the two things that raise it besides the clot itself		
What happens to right ventricular size and to right ventricular output		
What happens to the position of the interventricular septum		
What happens to left ventricular filling, left ventricular output, and systemic arterial pressure		
What happens to right coronary perfusion while right ventricular wall stress stays high		
What happens to right ventricular contraction once that perfusion falls, and what this means for a patient who is already hypotensive		
What happens to the arterial oxygen tension and to the arterial carbon dioxide tension, and which regions of lung explain each		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C05-BD1

Draw the upright lung and its three zones

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C05 Activity: NPLEX-P1-CARD-15-01-C05-BD1 Date: _____

Draw an upright lung with the pulmonic valve marked, and write the vertical distance from the valve to the apex and to the base. Write how much pressure a centimetre of blood column adds and how many centimetres of water make 1 mmHg. With a pulmonary artery pressure of 25/8 mmHg, calculate and write the systolic and diastolic pressure at the apex and at the base, and write what this does to flow per unit volume from apex to base. Then write the third pressure that acts on the alveolar wall capillaries and what it stays near in quiet breathing. Build a table of the three zones with one row each and the columns pressure order, blood flow, driving pressure, and usual location, and fill every cell. Beside the zone 2 row write why venous pressure does not affect flow there. Beside the zone 1 row write why it is normally absent.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C05-BD2

Make zone 1 appear, then work the ventilation and perfusion ratios

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C05 Activity: NPLEX-P1-CARD-15-01-C05-BD2 Date: _____

Write the two ways zone 1 can appear and give one clinical cause of each, then show with numbers why a positive end-expiratory pressure of 15 cmH2O creates zone 1 at the apex. Write what zone 1 alveoli do to carbon dioxide elimination and what has to happen to prevent the arterial carbon dioxide tension from rising, and write what exercise does to the zones instead. Next, write the apical and basal pleural pressures at functional residual capacity, explain from the pressure-volume curve why apical alveoli are more expanded but ventilate less per breath, and write whether ventilation or perfusion rises more steeply toward the base. Give the whole lung average ventilation-perfusion ratio with the two numbers it comes from, then give the ratio, the alveolar oxygen tension, and the alveolar carbon dioxide tension at the apex and at the base. Use the apical value to explain where reactivation tuberculosis sits and why. Then explain, using the shape of the oxyhaemoglobin dissociation curve, why high ratio regions cannot compensate for low ratio regions in oxygen but can for carbon dioxide, and write the blood gas pattern that follows, the normal alveolar to arterial oxygen difference in a young adult, and the two things that produce it. Finish by writing what happens to the apex to base differences when the patient lies down and which regions then receive most of the flow.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-15-01-C06-BD1

Shunt, and the work the pulmonary endothelium does

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C06 Activity: NPLEX-P1-CARD-15-01-C06-BD1 Date: _____

Write the definition of a shunt and the ventilation-perfusion ratio that goes with it. List the anatomical shunts, including the normal one and the share of cardiac output it carries, then list the intrapulmonary shunts with the conditions that cause them. Separately, write what the pulmonary endothelium does to angiotensin I, name the enzyme, the metal it needs, and the second name it carries, and write which peptide it inactivates and what accumulates when the enzyme is blocked and which two drug effects that explains. Write which mediators the lung clears in one passage and roughly how much norepinephrine it removes, and which three hormones pass through largely unchanged. Finish with the filter function: what the capillaries trap, the two reasons small emboli rarely infarct the lung, the condition that makes infarction more likely and why, and the name for a clot that crosses a patent foramen ovale along with how often that opening is present.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-15-01-C06-BD2

Separate the causes of hypoxaemia with oxygen and arithmetic

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-15 Heart and Lung Interactions Competency: NPLEX-P1-CARD-15-01-C06 Activity: NPLEX-P1-CARD-15-01-C06-BD2 Date: _____

Five hypoxaemic patients are breathing room air: one at 4,000 m, one after an opioid overdose, one with interstitial fibrosis, one with a COPD exacerbation, and one with dense right lower lobe pneumonia. For each write whether the alveolar to arterial oxygen difference is normal or raised, what the arterial carbon dioxide tension is doing, and whether 100% oxygen corrects the hypoxaemia. Then explain in full why oxygen barely helps the last patient: write what shunted blood never meets, the saturation the blood leaving ventilated alveoli already carries, how much oxygen dissolves per mmHg, and why mixing has to be done on content rather than on partial pressure before the dissociation curve turns that content into a low oxygen tension. Work the numbers for a 30% shunt with a haemoglobin of 15 g/dL, a saturated content of 20.1 mL/dL, and an arteriovenous difference of 5 mL/dL: give the arterial oxygen content, saturation, and oxygen tension on room air, then on 100% oxygen with an alveolar oxygen tension of 660 mmHg, and compare the final oxygen tension with a normal lung on 100% oxygen. Write why the arterial carbon dioxide tension stays normal or low. Finish with a child who has a ventricular septal defect and no cyanosis: write why there is no cyanosis at first, what the extra flow and pressure do to the pulmonary vessels, the six steps that end in reversal of the shunt, the name of that end state, the three clinical findings it brings, and why the vessels no longer relax by then.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-15-01-C06-CP02

Mechanisms of arterial hypoxaemia

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Mechanism	Alveolar to arterial oxygen difference	Arterial carbon dioxide tension	Response to 100% oxygen	Example condition
Low inspired oxygen tension				
Hypoventilation				
Diffusion limitation				
Low ventilation-perfusion regions				
Shunt with a ventilation-perfusion ratio of zero				
Alveolar dead space with a very high ventilation-perfusion ratio				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-15-01-C06-CA01

Hypoxaemia that does not lift on oxygen

A 64-year-old man is admitted with 3 days of fever, a productive cough, and worsening breathlessness. Temperature is 39.1 °C (102.4 °F), pulse is 112/min, respirations are 30/min, and blood pressure is 118/72 mm Hg. Breath sounds are reduced over the right lower chest with bronchial breathing and crackles, and the chest film shows dense consolidation of the right lower lobe with a clear left lung and a normal heart size. On room air, arterial blood gas analysis shows a PaO2 of 54 mm Hg, a PaCO2 of 31 mm Hg, and a pH of 7.47. He is placed on a non-rebreather mask delivering close to 100% oxygen, and a repeat sample 20 minutes later shows a PaO2 of 71 mm Hg. Mean pulmonary artery pressure measured during the admission is 18 mm Hg and the pulmonary capillary wedge pressure is 9 mm Hg.

1. Recognize. Name the mechanism of this man's hypoxaemia, give the ventilation-perfusion ratio of the affected lung units, and name the two features of his course that point to it.

2. Explain. Explain why oxygen content rather than oxygen tension has to be used when the two streams of blood mix, and why his arterial oxygen tension stays low once they have mixed.

3. Differentiate. Write how his blood gases and his response to oxygen differ from those of a patient with an opioid overdose and from those of a patient with a COPD exacerbation, naming the measurement that separates each one.

4. Investigate. Write which measurements you would use to show that the consolidated lobe rather than the left heart is the source, and the value or direction you would expect for each.

5. Decide. State what the local pulmonary arterioles are doing to the blood flow of the consolidated lobe and what his arterial oxygen tension would be without that response.

6. Justify. Justify that answer from the stimulus for the response, the channel events behind it, and the reason his mean pulmonary artery pressure stays near normal even though a whole lobe is constricted.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-15-01-C01-Q01

Why pulmonary pressure barely rises in exercise

A healthy 26-year-old woman is studied during heavy exercise. Her cardiac output rises from 5 L/min to 15 L/min, while mean pulmonary artery pressure rises only from 15 mm Hg to 22 mm Hg and left atrial pressure rises from 5 mm Hg to 10 mm Hg. Which of the following best explains the small rise in pulmonary artery pressure?

A. Sympathetic withdrawal relaxing the smooth muscle of the pulmonary arterioles

B. Diversion of part of the cardiac output into the bronchial circulation

C. Opening of previously unperfused capillaries together with widening of those already open

D. A rise in alveolar pressure that compresses the capillaries of the alveolar walls

Why did you pick it?

Board-style question NPLEX-P1-CARD-15-01-C04-Q02

Alveolar flooding at a normal wedge pressure

A 58-year-old man develops severe breathlessness 2 days after an episode of gram-negative sepsis. Respirations are 34/min and oxygen saturation is 84% on high-flow oxygen. The chest film shows diffuse bilateral alveolar opacities with a normal heart size. Pulmonary capillary wedge pressure is 8 mm Hg, and fluid suctioned from the airway has a protein concentration close to that of plasma. Which of the following best explains the pulmonary edema in this patient?

A. Injury to the alveolar capillary barrier that allows protein to cross at a normal capillary pressure

B. Backward transmission of a raised left atrial pressure through the pulmonary veins

C. A fall in plasma oncotic pressure that narrows the usual safety margin

D. Obstruction of pulmonary lymphatic drainage by the septic process

Why did you pick it?

Board-style question NPLEX-P1-CARD-15-01-C05-Q03

Dead space created at the apex

A 45-year-old woman is receiving mechanical ventilation after major blood loss. Positive end-expiratory pressure is increased to 15 cmH2O, which is about 11 mm Hg. Her pulmonary artery pressure is 18/6 mm Hg, and the apex of her lung sits about 20 cm above the pulmonic valve. Minute ventilation is held constant, and her arterial PCO2 rises over the next hour. Which of the following best explains the rise in arterial PCO2?

A. Hypoxic constriction of the arterioles supplying the apical alveoli

B. Loss of radial traction on the extra-alveolar vessels of the apex

C. Collapse of apical alveoli with continued perfusion of their capillaries

D. Alveolar pressure at the apex now exceeding arterial pressure, so those ventilated alveoli carry no blood flow

Why did you pick it?

Board-style question NPLEX-P1-CARD-15-01-C03-Q04

A large inspiratory fall in systolic pressure

A 51-year-old woman with metastatic breast cancer comes to the emergency department because of 2 days of breathlessness. Pulse is 118/min, blood pressure is 96/70 mm Hg, and the jugular veins are distended. During quiet breathing the systolic pressure falls by 22 mm Hg with each inspiration. Heart sounds are distant, the lungs are clear, and echocardiography shows a large circumferential pericardial effusion with equal diastolic pressures in the two ventricles. Which of the following best explains the inspiratory fall in systolic pressure?

A. Large swings in pleural pressure from obstructed expiratory airflow raise left ventricular afterload

B. Pericardial fluid fixes the combined ventricular volume, so inspiratory right ventricular filling displaces the septum into the left ventricle

C. Inspiratory constriction of the pulmonary arterioles reduces the volume returning to the left atrium

D. A rigid pericardium shields the heart from the fall in pleural pressure

Why did you pick it?